

Syllabus

§1 Day 1

1. Naive Set Theory: disjoint union
2. Equivalence relations
3. Canonical decomposition of a map
4. Definition of category
5. Isomorphisms
6. Initial and terminal objects
7. A commutative diagram
8. No HW.

§2 Day 2

1. What is a group?
2. Examples of groups
3. No HW.

§3 Day 3

1. Order of an element
2. Examples of order of elements
3. Property: if $g^n = e$, then order of g divides n .
4. symmetric groups (automorphism of set of 1 to n , $n!$, S_3)
5. dihedral groups D_{2n} (group element is operation)
6. cyclic groups ($\mathbb{Z}/n\mathbb{Z}$, who is the generator, order n)
7. units group $(\mathbb{Z}/n\mathbb{Z})^\times$, is there a generator? that is, is it cyclic? (this is very hard)
- 8.
9. HW: reading chapter 2 sections 1 and 2. Do 1.3, 1.4, 1.5, 1.8, 1.10, 1.11. Do 2.2, 2.5, 2.6, 2.7, 2.11, 2.12, 2.13, 2.15, 2.17

§4 Day 4

1. Group homomorphisms. $\phi(a * b) = \phi(a) \times \phi(b)$. Show $\phi(e) = e'$, $\phi(a^{-1}) = \phi(a)^{-1}$
2. Example: e as initial and final. $\mathbb{Z}$ to $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Z}$ to G by g^n . $(\mathbb{R}, +)$ to $(\mathbb{R}^\times, \times)$, S_3 to D_6 , S_3 to C_2 . $\det(M)$ as homomorphism from $GL_n(\mathbb{R})$ to $\mathbb{R}^\times$. $\mathbb{C}^\times$ to $\mathbb{R}^\times$ by $z \mapsto |z|$.
3. Category of groups.
4. Category of abelian groups.
5. Initial and final objects in groups: trivial group
6. Product groups: product is a universal object in some sense. What is product in sets? What is product in general. Product and coproduct in groups and abelian groups.
7. Some homomorphisms from and to product groups.
8. Group homomorphisms and order of elements.
9. Isomorphisms in general
10. groups: bijective homomorphisms (nontrivial, deserve a proof): $S_3 \cong D_6$, $\mathbb{Z}/6\mathbb{Z} \not\cong D_6$, $(\mathbb{Z}/7\mathbb{Z})^\times \cong \mathbb{Z}/6\mathbb{Z}$, $C_2 \times C_3 \cong C_6$.
11. homeomorphisms of topological spaces is not bijective morphism.
12. When not isomorphic? Key property, isomorphism preserves order of an element. Look at order of elements.
13. homomorphisms of abelian groups form a group, in fact, a ring.
14. automorphism of groups form a group.
15. HW: reading chapter 2, section 3, 4. Do 3.2, 3.4. Do 4.1, 4.3, 4.7, 4.8, 4.9, 4.10, 4.13, 4.14, 4.15, 4.17.

§5 Day 5

1. Free groups
2. Universal property of free groups, free groups is universal so unique up to isomorphisms.
3. Construction of free groups (maybe skip)
4. Construction of free abelian groups